MOHAMED HESHAM KHATTAB

Mechatronics Engineer

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Transferable Iqama | Saudi Council of Engineers (SCE) Registered Engineer

Objective

Mechatronics Engineer specializing in mechanical design, manufacturing, and industrial automation. Experienced in designing and developing mechanical systems for robotic and automated applications using SolidWorks, with hands-on experience in manufacturing processes, PLC programming, and industrial control. Seeking to contribute to the design and development of innovative engineering systems that integrate mechanical and automation technologies.

Education

Mansoura University – Faculty of Engineering **(Oct. 2020 – June. 2025)**

- B.Sc. in Mechatronics Engineering.
- CGPA: 3.71 / 4.0 (Excellent) | Ranked 20th in Cohort.

Professional Experience

Mechatronics Engineer – New Egypt Gold **(Oct. 2025 – Sep. 2026)**

- Perform reverse engineering and mechanical design of machine components and spare parts, including modifications and improvements to existing designs, and supervise their fabrication.
- Design and modify machine control panels and automation systems, including control circuits, PCB design, and motor-driver programming, as well as troubleshooting and maintenance of control systems.
- Handle CAD/CAM build preparation for wax-based additive manufacturing, while performing preventive maintenance, troubleshooting, and quality control to ensure reliable machine operation.

Professional Certifications

Saudi Council of Engineers (SCE) - Professional Accreditation **(Aug. 2026)**

- Professionally registered and accredited by the Saudi Council of Engineers as a Mechatronics Engineer.

Projects

1. Remotely Operated Underwater Vehicle – Graduation Project

- Designed the structure in SolidWorks, including material selection, dimensions and thruster configuration, achieving optimal performance and stable maneuvering.
- Designed a 3-DOF robotic arm, selecting MG996R servo motors for optimal torque and precision, integrating rotary position feedback for accurate control.
- Engineered a mechanical “Sport Mode” system for adaptive thruster orientation, improving propulsion efficiency by 79% and speed by 34% during navigation and maneuvering.

2. AI-Based Autonomous Research Boat

- Designed the internal structure using metal square tubes with sheet covers, ensuring stability and durability.
- Developed a buoyancy system using sealed PVC pipes for air storage, improving overall boat flotation.
- Designed a waterproof electronics box, ensuring protection with easy access for maintenance and integration

3. Heavy Weight Combat Mobile Robot (50 kg)

- Designed the robot’s weapon and power transmission, and selected suitable actuators for operation.
- Optimized the robot’s exterior to minimize incoming hits while maximizing weapon reach.
- Balanced weight distribution to comply with competition requirements, enhancing stability and performance.

Publications

Design and Development of Drivetrain System for Electric Motorcycle **(Nov. 2023 – Dec. 2024)**

- Presented at 25th International Middle East Power Systems Conference (MEPCON),
- Published on **IEEE Xplore.**
- Performed dynamic inertia analysis and mechanical simulation using SolidWorks Motion Analysis

Skills

- **Mechanical Design:** Machines Design, Mechanisms, DFM/DFA, Actuators integration, Reverse Engineering.
- **CAD & Simulation:** SolidWorks (Part, Assembly, Sheet Metal, Weldments, Drawings), Static Stress Simulation.
- **Manufacturing & Prototyping:** Additive Manufacturing, Sheet Metal Fabrication, CNC Machining.
- **Industrial Automation:** PLC Programming (TIA Portal), Classic Control, Motor Drives, Sensors & Actuators.
- **Microcontrollers & PCB Design**: Arduino, ESP32, AVR, Proteus, Schematic Capture, PCB Layout.
- **Soft Skills:** Teamwork, Adaptability, Fast learner, Project management, Communication, & Problem solving.

EXTRACURRICULAR ACTIVITIES

Momentum Team

Mechanical Design Committee Head **(Sep. 2023 – Sep. 2024)**

- Led the Mechanical Design Committee, managing recruitment, evaluations, and interviews.
- Delivered training on SolidWorks and mechanical components such as power transmission and shaft design.
- Taught key manufacturing methods, motor types, and motor selection based on load requirements.

Vice Technical Director **(Sep. 2023 – Sep. 2024)**

- Served as Deputy Technical Manager overseeing team progress in the manager's absence.
- Prepared weekly reports on each committee's performance and development.

IEEE Mansoura Student Branch

Technical Team Member in Micromuse Competition at IEEE Victoris 3.0 **(Aug. 2024 – Sep. 2024)**

- Provided continuous technical support to participating teams throughout the competition.
- Participated in judging and filtration and ensured all robots met technical requirements.

Multimedia Committee Head **(Oct. 2022 – Sep. 2023)**

- Restructured the Media Committee into specialized sub-teams to improve workflow.
- Worked with the Marketing Committee to develop and execute media campaigns for major events.

Graphic Design Coordinator at IEEE VICTORIS 1.0 **(June 2022 – Sep 2022)**

- Designed the overall competition model and defined all structural and logistical details.
- Defined technical session content, helped select instructors, and created the project evaluation scoring sheet.

Awards & Competitions

- **1st** Place **AAKRUTI Innovation Competition 2025,** Egypt Finale. **(Oct. 2025)**
- **1st** Place **IoT-Preneur Quest competition for Graduation Projects,** IEEE Victoris 4.0. **(Sep. 2025)**
- **6th** Place **International ROV Competition (ICMCT),** Military Technical College – UMVC. **(July. 2025)**
- **4th** Place **Metal Monsters Competition,** Alamein Robotics Championship. **(Oct. 2023)**
- **3rd** Place **Autonomous Sumo Competition,** IEEE HTI. **(Sep. 2023)**
- **1st** Place **Mini Metal Monsters Competition,** Maker Faire Cairo. **(Oct. 2022)**

Language

- **Arabic :** Native.
- **English :** Very Good (Professional Working Proficiency).